1. Implement A* Star Search algorithm with Manhattan distance.

   Output:

```
Start:
2 8 3
1 6 4
7 _ 5

Goal:
1 2 3
8 _ 4
7 6 5

Iter 1: expand 283164705 (g=0, h=5, f=5)
    Up    -> 283104765 (g=1, h=4, f=5)
    Left  -> 283164075 (g=1, h=6, f=7)
    Right -> 283164750 (g=1, h=6, f=7)
Iter 2: expand 283104765 (g=1, h=4, f=5)
    Up    -> 203184765 (g=2, h=3, f=5)
    Left  -> 283014765 (g=2, h=5, f=7)
    Right -> 283140765 (g=2, h=5, f=7)
Iter 3: expand 203184765 (g=2, h=3, f=5)
    Left  -> 023184765 (g=3, h=2, f=5)
    Right -> 230184765 (g=3, h=4, f=7)
Iter 4: expand 023184765 (g=3, h=2, f=5)
    Down  -> 123084765 (g=4, h=1, f=5)
Iter 5: expand 123084765 (g=4, h=1, f=5)
    Down  -> 123784065 (g=5, h=2, f=7)
    Right -> 123804765 (g=5, h=0, f=5)
Iter 6: expand 123804765 (g=5, h=0, f=5)
```

```
Solution path:
Step 1: Up
2 8 3
1 _ 4
7 6 5

Step 2: Up
2 _ 3
1 8 4
7 6 5

Step 3: Left
_ 2 3
1 8 4
7 6 5

Step 4: Down
1 2 3
_ 8 4
7 6 5

Step 5: Right
1 2 3
8 _ 4
7 6 5

Solved in 5 moves, 6 iterations.
Moves: Up -> Up -> Left -> Down -> Right
```

2. Implement A* algorithm with misplaced tiles.

   Output:

```
Start state:
2 8 3
1 6 4
7 _ 5

Goal state:
1 2 3
8 _ 4
7 6 5

Step 1: Up  (g=1, h=3, f=4)
2 8 3
1 _ 4
7 6 5

Step 2: Up  (g=2, h=3, f=5)
2 _ 3
1 8 4
7 6 5

Step 3: Left  (g=3, h=2, f=5)
_ 2 3
1 8 4
7 6 5

Step 4: Down  (g=4, h=1, f=5)
1 2 3
_ 8 4
7 6 5

Step 5: Right  (g=5, h=0, f=5)
1 2 3
8 _ 4
7 6 5

Solved in 5 moves (6 nodes expanded).
Moves: Up -> Up -> Left -> Down -> Right
```